Preliminary Design Report

A Multi-Agent Robotic System for Autonomous Ice-Shell Characterization and Cryobot Site Selection on Europa

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Space Robotics — 4.1 Preliminary Design Report
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GAP — Needs Author Input

This report is a first-pass synthesis of every prior deliverable in this repository into the Section 4.1 template. It was compiled and drafted with the assistance of Claude (Anthropic), which read the existing assignments (1.1 through 3.6) and assembled, reorganized, and lightly rephrased that material into the structure the assignment requires; it did not invent new mission design decisions. Every technical number, requirement, trade study, and calculation below originates in the author's own prior work product in this repository unless a gap box (like this one) says otherwise. Gap boxes throughout the report mark: (1) content that has no completed source assignment yet (Command & Data Handling design, the flight software state diagram, the full requirements-verification status), and (2) numbers that were reconstructed or paraphrased from adjacent documents rather than copied verbatim, and should be checked against the master spreadsheet before submission. A consolidated gap list is repeated at the end of the document. See also Appendix D for the itemized AI-use disclosure carried over from each source assignment.

Contents

1 Executive Summary

This mission deploys a coordinated multi-agent robotic system to Europa, Jupiter's icy moon, to autonomously characterize ice-shell thickness along a surface traverse and identify an optimal cryobot insertion site, operating without real-time Earth support [1]. The system consists of a surface rover carrying a ground-penetrating radar sounder, supported by two orbital relay agents that provide terrain reconnaissance, communications relay, and cooperative localization [3]. This mission directly addresses three converging research gaps: autonomous operations in GPS-denied, high-radiation environments; multi-agent coordination and interoperability for cooperative task planning [2]; and fully autonomous approach and proximity operations without Earth-based navigational support [3].

Europa was chosen as the destination because it holds strong indicators of conditions favorable to life and contains the chemical building blocks necessary to support it [1]. Its subsurface ocean makes it one of the most promising locations in the Solar System for astrobiology, but the 44-minute one-way light delay to Jupiter, Jupiter's intense radiation environment, and a 90 K cryogenic surface make it one of the most demanding environments to operate a robot in. Every decision from landing onward must be made onboard; there is no real-time human-in-the-loop safety net.

The rover is a four-wheel, differential-drive vehicle riding on a single-pivot rocker, sized against a 147.25 kg contingency mass cap, and powered by a Radioisotope Thermoelectric Generator (RTG) rather than solar arrays, since solar irradiance at Europa's distance from the Sun collapses to roughly 50 W/m^2. Its primary payload is a REASON-heritage dual-frequency, ice-penetrating radar sounder, the same instrument family currently flying aboard NASA's Europa Clipper spacecraft [7]. Where Clipper collects wide-swath, low-resolution orbital snapshots of the ice shell, this mission's rover drives a continuous surface traverse, building a spatially registered subsurface profile at 100 m resolution or better and autonomously flagging the thinnest viable ice for a future cryobot deployment.

Two orbital agents support the rover from a 200 km circular Europa orbit, providing terrain mapping ahead of the traverse, a UHF proximity relay for the rover's science and command traffic, and an independent ranging source the rover can use to recover its position after a localization fault. Communication with Earth exists only as a low-rate contingency beacon; a direct link at Europa's distance closes at roughly 7 bits per second, which is why the relay architecture is not optional but foundational to the mission design.

Across a 30-sol surface mission, the system is expected to produce a continuous, georeferenced subsurface ice-thickness profile along the traverse track, a validated candidate cryobot insertion site (ice thickness under 5 km, surface slope under 20°, terrain roughness under 0.7), and a full science and engineering telemetry archive suitable for both real-time onboard decision-making and post-mission analysis by Earth-based planetary scientists. The requirements, environmental analysis, concept of operations, payload rationale, data strategy, and subsystem designs that support this objective are detailed in the sections that follow.

2 Research Context and Gap Framing

Three converging research gaps motivate this mission, drawn from the OWL Planetary Decadal Survey, a NASA STMD technology shortfall, and a NASA SBIR/STTR solicitation topic.

2.1 OWL Decadal Survey — Ocean Worlds Exploration & Sampling

Topic: Ocean Worlds Exploration & Sampling [1, pp. 36–39]

Interest: Ocean worlds like Europa represent the most compelling environments for studying life beyond Earth, and the most difficult for autonomous systems. What draws me to this topic is the engineering challenge that arises from the question: how do you keep a robot operational, navigating, and decision-making in a GPS-denied, high-radiation, cryogenic environment with no possibility of real-time human intervention? The decadal survey speaks to the real physical needs here: Jupiter's Galilean moons reveal "the fundamental importance of gravitational tides in generating internal heat," with those tides actively "melting ice on Europa and Ganymede" [1, p. 37]. That subsurface ocean exists because of a process we cannot turn off, and any robotic system sent to study it must be designed around that reality.

Connection to Space Exploration: Accessing a subsurface ocean requires penetrating kilometers of ice, a process that may yield in-situ resources (liquid water, chemical energy gradients) critical to sustaining any long-duration mission. Understanding the composition of that ocean directly informs the feasibility of using local materials to support future human or robotic infrastructure in the outer solar system.

Relationship to Robotics: Reaching and operating in a subsurface ocean demands a full robotic stack operating without outside reference: specialized robots for ice penetration, underwater vehicles for ocean navigation, and autonomous sampling systems capable of detecting and responding to scientifically interesting features. Every subsystem (perception, localization, path planning, controls) must function independently.

2.2 NASA STMD Shortfall — Multi-Agent Robotic Coordination

Topic: Multi-Agent Robotic Coordination and Interoperability for Cooperative Task Planning and Performance [2]

Interest: Swarm robotics is the next step in advancing the capabilities of robots to complete high-level missions. NASA's own shortfall rankings confirm how urgent this is: multi-agent coordination received an integrated score of 4.7856 and ranked 139th among all civil space technology gaps [2], placing it right in the near-term priority tier for Autonomous Systems and Robotics. I am most interested in the version of it where each robot in the swarm is genuinely capable on its own, not a simple unit that only functions within the group, but a system of robots with unique capabilities that all come together to complete a high-level task.

Connection to Space Exploration: Cooperative exploration and mapping by a distributed robotic fleet dramatically increases the spatial coverage and fault tolerance of resource missions. A swarm can survey a large asteroid surface, map subsurface ice deposits via distributed sensing, or maintain constant monitoring of a volatile-rich region far more efficiently than any single agent, with adversarial identification algorithms when individual units fail.

Relationship to Robotics: This topic is a direct application of the consensus and coordination algorithms I have studied for the last year. Deploying them in space introduces new constraints (limited bandwidth, high latency, radiation-induced faults, and no GPS) that stress-test the theoretical properties I've studied (convergence, resilience, scalability).

2.3 NASA SBIR/STTR Topic — Flight Dynamics and Navigation Technologies

Topic: Flight Dynamics and Navigation Technologies [3] (Topic COMNAV.1.S26B)

Interest: Communication delays and blackouts are not random edge cases in deep-space missions; they are the default operating condition. The SBIR solicitation states the gap directly: "NASA currently has limited navigational and guidance technologies that permit fully autonomous approach, proximity operations, distributed spacecraft operations, and landing without support from Earth-

based resources" [3]. This shortfall interests me because it forces onboard autonomy to be the primary decision-making layer, eliminating the safety net of human-in-the-loop oversight.
Connection to Space Exploration: Remote resource operations (prospecting, extraction, processing) cannot depend on continuous communication to Earth. Whether managing an ISRU plant on the lunar surface or conducting a sampling traverse on an asteroid, the system must detect anomalies, replan, and execute without waiting for human feedback.
Relationship to Robotics: My research in consensus algorithms and resilient multi-agent coordination already deals with how distributed systems reach agreement under adversarial conditions. Extending this to communication-constrained space contexts is a natural progression: the same algorithms that allow a swarm of Crazyflies to maintain stability during packet loss should also apply to a team of planetary rovers operating through a communication outage.

3 Mission Definition and Environment

3.1 Mission Objective & Destination

Selected Research Gap: While most robotic space missions have focused on a single agent carrying all the capabilities needed to complete the mission objectives, the future of advanced planetary exploration lies in multi-agent systems that are able to share responsibility and support each other. In order to fully understand a body as complex as Europa, it is necessary to divide surveying tasks across the main exploration rover and the orbital agents needed to scout terrain ahead of it. This mission directly addresses three converging research gaps: autonomous operations in GPS-denied, high-radiation environments [1]; multi-agent coordination and interoperability for cooperative task planning [2]; and fully autonomous approach and proximity operations without Earth-based navigational support [3].
Mission Statement: The multi-agent Europa surface system shall autonomously characterize the ice shell and identify a cryobot insertion site, operating without real-time Earth support.
Success Criteria (summary): Full traverse track covered at $\leq$ 100 m along-track spacing, ice-thickness accuracy within $\pm 10\%$ against calibration targets, and at least one candidate insertion site flagged by the onboard algorithm (ice thickness $<$ 5 km, surface slope $< 20°$, terrain roughness $<$ 0.7). Section 3.3 develops the full mission- and operations-level requirements that this criterion is built from.
Destination Justification: My mission takes place on Europa, Jupiter's icy moon. I chose this destination due to the abundance of knowledge there is to gain from an in-depth exploration of the system. Europa holds strong indicators of conditions favorable to life and contains all the chemical building blocks necessary to support it [1]. The layers of ice introduce in-situ resources that might benefit further exploration of the moon as long-term infrastructure is developed, as well as imposing significant rover design constraints. Any surface vehicle must feature traction-optimized wheels capable of accurately traversing the fractured, chaotic terrain.

Europa is also incredibly challenging. The 44-minute one-way light delay means every decision from landing onward must be fully autonomous [3]. Due to the intense radiation environment driven by Jupiter's magnetosphere, the rover relies on orbiting satellite robots to scan the terrain and share terrain maps prior to surface traversal. Such a system has to be highly robust to faulted or adversarial agents, as incorrect or incomplete maps shared across the swarm are likely to produce dangerous navigation decisions. This location is essential to the mission because it provides an incredibly challenging environment that necessitates smart, advanced robot design and algorithm development — the perfect stage to push robotic capabilities to their limits.

3.2 Environmental & Operational Challenges

Europa's environment is hostile in ways that shape every design decision in this mission. Jupiter's magnetosphere floods the moon's surface with high-energy charged particles, exposing electronics to radiation doses far greater than what commercial components can tolerate. This motivates a requirement for radiation-hardened electronics and a shielded vault setup for the rover's core computing hardware, similar to the approach taken on the Galileo spacecraft [1]. Redundancy at the agent level is an equally important mitigation: if one surface relay node builds up enough single-event upsets to go silent, the remaining agents must autonomously reconfigure the network to maintain coverage.

Surface temperatures on Europa are around 90 K, which introduces thermal management challenges for both batteries and actuators. Power storage capacity drops significantly at cryogenic temperatures, meaning the rover must actively manage its thermal systems to keep cells within an operational range rather than simply relying on passive insulation. Solar power is not viable at Europa's distance from the Sun (irradiance collapses to roughly 50 W/m^2), so the system instead relies on an RTG.

The complete absence of GPS and the communication blackout imposed by the 44-minute light delay mean the swarm cannot depend on any external reference for localization or replanning. Each agent must maintain its own state estimate using onboard sensors and inter-agent ranging, and the swarm must reach consensus on shared map data using algorithms that are resilient to packet loss and radiation-induced faults [2, 3]. There is no human in the loop to catch a bad map before the rover commits to a traverse. Table 1 summarizes how each environmental driver maps onto a subsystem-level design response; the full derivation appears in the subsystem sections of Section 7.

Environmental Driver	**Operational Consequence**	**Design Response**
Jupiter magnetosphere radiation ($\geq$ 100 krad)	Electronics degradation, single-event upsets	Radiation vault (Ti-6Al-4V + Pb/poly liner), agent-level redundancy
44-min one-way light delay	No real-time human command	Full onboard autonomy, consensus-based multi-agent decisions
No GPS, no magnetic field	No absolute position/heading reference	Visual odometry + IMU + fine sun sensors + inter-agent ranging
90 K cryogenic surface	Battery capacity loss, actuator lubricant stiffening, brittle materials	MLI + RTG waste-heat routing + survival heaters, pre-motion warm-up
$\sim$ 50 W/m^2 solar irradiance	Solar arrays not viable	RTG primary power source
Fractured, chaotic ice terrain	Traction/mobility risk, hazard density	Cleated wheels, onboard hazard rejection, low-speed arc-based traversal

Table 1: Environmental drivers and their design responses.

3.3 Mission Success Criteria

Requirements are drawn from the master requirements spreadsheet (Appendix B contains the full set, REQ.01.01 through REQ.11.04). Table 2 presents three mission-level requirements and Table 3 presents three operations-level requirements, each with its justification, verification approach, and associated risk.

ID	Statement	Justification	Verification / Risk
REQ.01.01	The multi-agent Europa system shall operate fully autonomously for the entire surface mission duration without requiring real-time command uplinks from Earth.	The 44-minute one-way light delay makes real-time commanding physically impossible; all decisions must be onboard.	72-hr hardware-in-the-loop mission sequence with a simulated blackout, zero unresolved Earth dependencies. Risk-01: single-point autonomy failure with no fallback human override.
REQ.01.02	The system shall survive and maintain operational status after exposure to a total ionizing dose of no less than 100 krad over the nominal mission lifetime.	Jupiter's magnetosphere delivers doses far exceeding near-Earth missions; Europa Clipper's radiation vault targets ~ 100 krad as its design threshold.	Co-60 gamma irradiation of radiation-sensitive components to 110 krad (10% margin) with functional monitoring throughout. RISK-02: accumulated SEUs degrade consensus convergence before mission completion.
REQ.01.03	The system shall maintain a minimum inter-agent network connectivity of two active relay nodes at all times during surface operations to ensure consensus is achievable.	Consensus algorithms require a minimum connected graph; two relays plus the rover guarantee a 3-node connected topology, the minimum viable configuration.	Network fault simulation with progressive relay failures; reconfiguration must trigger before dropping below 2 active nodes. RISK-03: simultaneous radiation-induced failure of multiple relay nodes isolating the rover.

Table 2: Mission-level requirements.

ID	Statement	Justification	Verification / Risk
REQ.02.01	The surface robot shall maintain a localization accuracy of no worse than ±0.5 m relative to its last known position during traverse segments up to 50 m in the absence of GPS.	Cryobot insertion site selection requires meter-level positioning to match the radar-identified thin-ice candidate; consistent with visual odometry performance on Mars rovers.	Analog-terrain field test, 50 m commanded segments, onboard estimate vs. ground truth across 20 runs. RISK-04: ice surface texture homogeneity reducing visual odometry feature tracking.
REQ.03.01	The ground-penetrating radar sounder shall resolve ice shell thickness to within ±10% accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at a minimum spatial resolution of 100 m along the traverse track.	Identifying the optimal cryobot insertion site requires knowing where the ice shell is thinnest; consistent with Europa Clipper REASON design parameters.	Laboratory radar range measurements against known-thickness ice-analog targets across the full depth range. RISK-05: ice-shell heterogeneity or brine inclusions creating radar clutter.
REQ.04.01	The inter-agent mesh network shall achieve consensus on a shared terrain map update within no more than 10 minutes under a degraded network condition of up to 30% packet loss on any single inter-agent link.	Traverse planning depends on a current, consensus-validated terrain map; a 10-minute bound keeps planning delay acceptable, and 30% loss is a plausible radiation-induced degradation scenario.	Software-in-the-loop consensus simulation, 30% packet loss injected on every link across 100 test runs. RISK-06: radiation-induced burst errors exceeding 30% loss on multiple links simultaneously, stalling consensus.

Table 3: Operations-level requirements.

GAP — Needs Author Input

The master spreadsheet marks every requirement's *Status* field as "Unaccomplished" as of the last update (6/14/2026–8/6/2026). Before final submission, revisit the V&V columns in the master spreadsheet and update status/notes for whichever requirements have since been verified by analysis, test, or demonstration this semester, and carry that status into Appendix B.

4 Concept of Operations (ConOps)

4.1 Overview Narrative

The multi-agent system deployed to Europa's surface autonomously characterizes ice shell thickness and identifies an optimal cryobot insertion site without real-time Earth support. The destination is Europa, chosen because its subsurface ocean makes it one of the most promising locations in the Solar System for astrobiology and future exploration. The mission deploys a surface rover equipped with a ground-penetrating radar sounder, supported by two orbital reconnaissance agents that provide terrain mapping, communications relay, and cooperative localization. By leveraging radar sounding techniques analogous to the Europa Clipper REASON instrument, the system generates detailed subsurface ice maps to support future cryobot missions.

The mission consists of six main phases: Launch and Commissioning, Cruise and Approach, Europa Orbit Insertion and Reconnaissance, Landing, Surface Checkout and Site Characterization, Surface Survey and Ice Characterization, and End-of-Mission Operations. Launch and cruise prepare the system for arrival at Europa. Upon arrival, the orbital agents conduct reconnaissance and identify a suitable landing region. Landing is followed by rover deployment, system checkout, and sensor calibration. Main surface operations focus on autonomous rover traversal while collecting radar measurements of the ice shell and building a subsurface map; the orbital agents continuously update terrain maps and relay communications. The final phase includes science data downlink, candidate cryobot site selection, optional extended operations, and safe system passivation.

The rover operates autonomously during critical events such as landing, navigation, radar sounding, and site-selection activities. The two orbital agents provide terrain reconnaissance, communications relay, cooperative localization support, and validation of shared mapping data. Ground teams remain in the loop during cruise, strategic mission planning, and anomaly resolution, but all tactical decisions must be performed onboard due to the approximately 43–44 minute one-way communication delay between Earth and Jupiter. Radiation exposure, cryogenic temperatures, communication delays, and limited bandwidth are the key operational constraints that shape every phase below.

4.2 Operational Timeline

Table 4 presents the mission-phase-level ConOps timeline (mission phases, operation modes, and key events), and Figure 1 presents the corresponding operational flow graphic. A finer-grained, minute-by-minute breakdown for the first three sols of surface operations plus the steady-state daily cycle and end-of-mission sequence is maintained in the source spreadsheet (*2.4 ConOps Detailed Timeline*); it is summarized in Table 5 rather than reproduced row-by-row here for length.

Time	**Mode**	**Objective / Milestone**	**Op. Mode**	**Risk**	**Duration**
Launch day, 0–1	Launch & Commissioning	Spacecraft separation, initial health checks	Scripted/Auto	High	1 day
Cruise, yrs 0–6	Cruise & Approach	Maintain health, trajectory correction maneuvers	Semi-Auto	Medium	∼6 years
Europa arrival	Orbit Insertion & Reconnaissance	Deploy orbital agents, identify landing region	Autonomous	High	30 days
Surface day 0–5	Landing, Checkout & Site Characterization	Deploy rover, verify health, calibrate payload	Autonomous	High	5 days
Surface day 5–35	Surface Survey & Ice Characterization	Map subsurface ice, identify candidate cryobot sites	Autonomous	Medium	30 days
Surface day 35–50	Data Transmission & End of Mission	Downlink archive, select final site, passivate	Semi-Auto	Low	15 days

Table 4: Mission-phase-level Concept of Operations timeline.

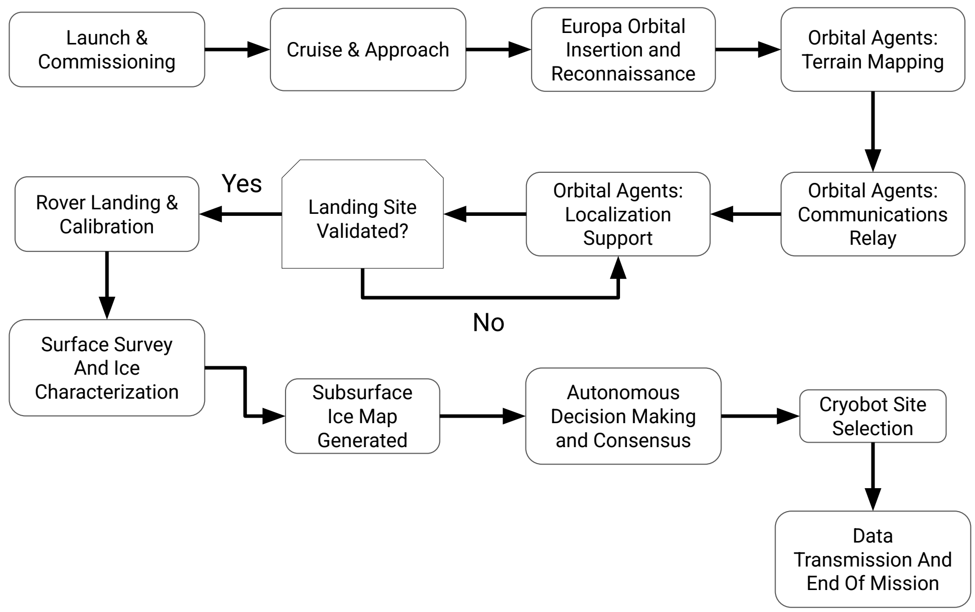

Figure 1: Operational flow across the six mission phases, including the landing-site validation decision point and the autonomous consensus loop that drives cryobot site selection.

Sol(s)	Phase	Key Events
0	Initialization	Descent/landing (0–30 min); commissioning and health checks; orbital comms verification; localization initialization; radar payload init, calibration, and validation; initial site characterization; night low-power thermal management.
1–2	Activation	Daily cycle established: autonomous traverse planning, orbital terrain mapping update, surface survey blocks, localization verification, radar data collection, communications relay pass, subsurface ice map generation.
3–25	Steady-state operations	Repeating daily cycle: traverse planning → survey operations → radar data collection → communications relay pass → subsurface ice map update → night low-power thermal management.
26–28	Site selection	Candidate site ranking and autonomous consensus validation (repeated over two sols), final site assessment, cryobot site selection.
29	Archive & validation	Final science archive generation, final terrain map validation.
30	End of mission	Final data dump, rover/payload safing, communications shutdown, passivation and final health beacon.

Table 5: Summary of the minute-by-minute surface operations timeline (source: 2.4 ConOps Detailed Timeline spreadsheet).

4.3 Flight Software State Diagram

GAP — Needs Author Input

No completed deliverable exists yet for a flight-software state machine (the template references a "2.5 ConOps and Flight Software" assignment, which is not present in this repository as of this draft — the closest existing work is the ConOps phase flowchart in Figure 1). Before submission, build an actual state diagram with clearly drawn states, transitions, and decision points. The mission phases and modes already defined across the ConOps and GNC deliverables give a natural starting set of candidate states: Launch/Commissioning, Cruise, Europa Orbit Insertion, Descent & Landing, Checkout & Calibration, Autonomous Traverse (with sub-modes *Drive Arc*, *Radar Sounding Stop*, *Hazard Replan*), Navigation Hold (entered per REQ.11.04 when 1σ localization uncertainty exceeds 0.5 m), Consensus / Site Selection, Data Downlink, Safe Mode (entered on TID, thermal, or power fault), and Passivation. The transition guards are largely already written as requirement thresholds (e.g., REQ.11.02's hazard-rejection criteria, REQ.11.04's navigation-hold entry/exit condition, REQ.05.03's power margin floor) and just need to be redrawn as a formal diagram (e.g., in draw.io, matching the house style of the other subsystem diagrams in this repository) and inserted here.

5 Payload-Driven Design

5.1 Key Payload Description

Payload: REASON (Radar for Europa Assessment and Sounding: Ocean to Near-surface)

REASON is a dual-frequency ice-penetrating radar sounder currently flying aboard NASA's Europa Clipper spacecraft. The instrument was developed by researchers at the University of Texas Institute for Geophysics and fabricated by JPL, and it launched on October 14, 2024 atop a SpaceX Falcon Heavy rocket bound for Europa [6]. REASON operates at two frequency bands simultaneously, allowing it to distinguish between near-surface structure and deep basal reflections, a capability no previous Europa-targeted instrument has demonstrated [4]. The instrument is flight-proven hardware with full mission-level heritage, placing it at TRL 9.

Europa Clipper carries REASON as part of a flyby science campaign: the spacecraft loops past Europa roughly 50 times, collecting wide-swath regional maps of the ice shell from altitude [5]. This mission seeks to address what Clipper does not cover: surface-level, high-resolution traverse profiling targeted at a specific candidate cryobot insertion site. There is no rover, no in-situ localization, and no onboard site-selection logic on Clipper, and the orbital geometry limits any single flyby to a brief snapshot rather than a continuous along-track profile. This mission is essentially a continuation of Clipper: a surface rover carrying an adapted REASON-heritage sounder drives a continuous traverse, building a spatially registered subsurface map at 100 m resolution or better, and autonomously flags the thinnest viable ice for a future cryobot deployment.

REASON measures four primary parameters. Two-Way Travel Time (ns) is the elapsed time between the transmitted pulse and the received basal reflection, and serves as the raw measurement. Signal Amplitude (dB) captures the strength of the return from the ice-ocean interface; echoes that fall into the noise floor are flagged to prevent false positives in site selection. Bulk Ice Dielectric Permittivity (dimensionless) is estimated from the sounding's travel characteristics ($\varepsilon_r \approx 3.15$ for pure water ice), with deviations flagging brine pockets or structural impurities. Finally, Derived Ice Shell Thickness (m) is computed autonomously onboard using $d = c \cdot t \,/\, (2\sqrt{\varepsilon_r})$, where c is the speed of light.

There are two crucial users of REASON's data. The onboard autonomous planning algorithm is the real-time consumer: it continuously processes daily trace summaries to locate viable cryobot deployment sites (ice thickness < 5 km, surface slope $< 20°$, roughness index < 0.7) and issues a deployment trigger without human intervention when conditions are satisfied. Earth-bound planetary scientists form the long-term consumer base: the full downlinked dataset provides a high-resolution, spatially registered along-track profile of Europa's ice shell that can be used to model global habitability, constrain ocean-access scenarios, and plan future deep-subsurface entry missions.

5.2 Payload Requirements

The payload imposes concrete, traceable requirements on every other subsystem. Pointing: REQ.03.02 requires the antenna boresight to stay within $\pm 5°$ of nadir during active sounding, forcing the rover to halt traversal whenever surface slope exceeds 20°; this requirement is carried by Structures (antenna mast/mounting geometry) and Mobility (slope-triggered halt logic). Power: REQ.03.03 requires a dedicated peak power allocation for radar transmission pulses and commands drive-motor shutdown during each sounding window, since the RTG's fixed continuous output cannot support simultaneous driving and sounding without risking under-powering the radar pulse. Data handling: REQ.03.04 requires the CDH subsystem to retain a minimum of 7 sols of uncompressed radar trace data and apply lossless compression at a minimum 2:1 ratio before inter-agent relay transmission, since raw science data accumulates faster than the non-deterministic Earth relay link can clear it. Thermal: REQ.05.01 requires the radar electronics (and all avionics) to stay within 0–40°C during active operations and above $-20°$C in night survival mode. Together these four requirements are why the payload, rather than a generic bus design, drove the rover's mobility cadence (Section 7.7), power budgeting (Section 7.4), onboard storage strategy (Section 6), and thermal architecture (Section 7.5).

6 Data Strategy and Deliverables

6.1 Data Product Table

Mission Objective: Autonomously characterize Europa's ice shell thickness along a surface traverse and identify an optimal cryobot insertion site without real-time Earth support (REQ.03.01: ground-penetrating radar sounder resolves ice thickness to $\pm 10\%$ for depths of 1–30 km at ≥ 100 m along-track spatial resolution).

The data product is organized into four sheets so the dataset stays clear and usable. Table 6 summarizes each; full field-by-field definitions live in the source 2.1 Data Product deliverable.

Sheet	**Key Fields (units)**	**Intended Consumers**
Radar_Traces	Trace ID, sol, along-track position (m), two-way travel time (ns), amplitude (dB), permittivity (–), derived ice thickness (m), quality flag	Onboard site-selection algorithm (real time); planetary scientists (post-mission)
Localization_Log	Position X/Y (m, local ENU), heading (deg), 1σ uncertainty (m), REQ.02.01 error flag	GNC/onboard planner; mission operations engineers
Terrain_and_Health	Surface slope (deg), roughness index (0–1), surface temperature (K), accumulated TID (krad), radar noise floor (dB), inter-agent packet loss (%), per-trace quality flag	Onboard hazard/site filter; systems engineers monitoring health trends
Daily Summary	Traverse distance/sol (m), valid trace count, minimum ice thickness/sol (m), candidate site count, mean consensus convergence time (min)	Mission planners (per-sol go/no-go); operations team

Table 6: Data product sheets, key fields, and intended consumers.

Each data category maps to a specific requirement. The radar sounder returns are the direct measurement required by REQ.03.01; without travel time and permittivity there is no thickness inversion and no subsurface map. The localization log is required by REQ.02.01: each trace must be placed within ±0.5 m to produce a spatially accurate map rather than an unordered list of depth readings. Terrain context drives the site-selection filter, where slope and roughness constrain physical accessibility independent of ice thickness. System health data is required by REQ.01.02 and REQ.04.01: elevated TID can degrade radar performance, and packet loss above 30% directly stresses the consensus convergence bound. The main known limitation is permittivity estimation uncertainty when brine inclusions or heterogeneous layering are present (RISK-05); brine lenses increase local permittivity and can cause the thickness algorithm to underestimate depth to the basal interface. The `CLUTTER` quality flag mitigates this by excluding affected traces from site selection, but does not resolve the ambiguity without a secondary instrument.

6.2 Example Output

On Sol 1 at 94.4 s elapsed mission time (trace `TR-0001`), REASON recorded a two-way travel time of 21,647.3 ns, a return amplitude of −41.62 dB, a bulk ice permittivity of 3.06, and a derived ice shell thickness of 1,855.5 m at rover position (93.8 m east, −7.1 m north), with an accumulated total ionizing dose of 0.542 krad and the record flagged as `NOMINAL`. Table 7 presents this as a sample row.

Trace ID	Sol	Pos. E (m)	Pos. N (m)	TWT (ns)	Ampl. (dB)	ε_r	Thickness (m)	TID (krad)	Flag
TR-0001	1	93.8	−7.1	21,647.3	−41.62	3.06	1,855.5	0.542	NOMINAL

Table 7: Example Radar_Traces record.

GAP — Needs Author Input

Only a single illustrative record (shown above) has been generated so far, and it was AI-generated to demonstrate the schema rather than sampled from an actual simulation or field test. Export a short run (10–20 consecutive rows) of the actual *Data Product Space Robotics* spreadsheet and drop it in here (and in Appendix C) before submission so the report shows real structured output rather than one synthetic row.

7 System and Subsystem Designs

7.1 Functional & Nonfunctional Requirements

Table 8 and Table 9 highlight five functional and five nonfunctional requirements drawn from the master requirements set (Appendix B has the complete REQ.01.01–REQ.11.04 list with sources, subsystem ownership, and V&V plans).

ID	Functional Requirement (what the system must do)
REQ.02.01	Maintain localization accuracy no worse than ± 0.5 m over any 50 m traverse segment in the absence of GPS.
REQ.03.01	Resolve ice shell thickness to within $\pm 10\%$ for 1–30 km depths, at ≥ 100 m along-track resolution.
REQ.06.01	Autonomously plan and execute ≥ 100 m of surface traverse per sol, pausing at ≤ 100 m intervals for a radar sounding stop, and resume without ground uplink.
REQ.11.02	Assess terrain out to ≥ 8 m ahead of each drive arc and autonomously reject any arc with a step > 0.20 m, slope $> 20°$, or roughness > 0.7.
REQ.11.04	Halt traverse and enter a navigation hold whenever position uncertainty exceeds 0.5 m (1σ), and re-initialize via relay ranging plus map correlation.

Table 8: Representative functional requirements.

ID	Nonfunctional Requirement (quality attribute / constraint)
REQ.01.01	Operate fully autonomously for the entire surface mission without real-time Earth uplinks (dependability / autonomy constraint).
REQ.01.02	Survive and remain operational after ≥ 100 krad total ionizing dose over the mission lifetime (survivability).
REQ.05.01	Maintain avionics/battery temperature within 0–40°C active, > -20°C night mode (environmental performance).
REQ.08.01	Generate ≥ 110 We (BOL) / ≥ 80 We continuous via MMRTG-class RTG, ≥ 1.9 kWh usable per sol (capacity / performance).
REQ.10.03	Close the rover–relay proximity link with ≥ 3 dB margin over the required E_b/N_0 at the 588 km worst-case range (reliability / performance margin).

Table 9: Representative nonfunctional requirements.

7.2 Robot System Overview

The system is a three-agent architecture: one surface rover and two orbital relay agents in 200 km circular Europa orbits. Broadly, the rover carries the science payload (REASON-heritage GPR sounder) and mobility/GNC stack needed to physically reach and characterize a candidate cryobot site; the orbital agents carry the terrain-mapping and communications-relay capability the rover cannot provide for itself in a GPS-denied environment. Together they fulfill the mission objective end to end: the orbiters find where it is safe to look, the rover determines exactly how thick the ice is there, and the mesh network gets that answer back to Earth despite a 44-minute light delay.

Figure 2 shows how the rover's subsystems interface with one another. Power (RTG + battery packs) feeds every other subsystem over regulated 28 V and 5 V buses through the PDU/EPS board; Thermal routes RTG waste heat to the radiators and survival heaters and isolates the radiator loop with MLI and thermal switches; Structures provides the physical mounting, radiation vault, and load path that the deployed legs, mast, and antenna assembly all attach to; Avionics (the RAD750-class flight computer and I/O boards) hosts the GNC compute stack and mediates between Comms, Payload, and Mobility; and Comms carries science data out over UHF to the relay agents and commands back in, with an X-band link reserved as a low-rate, direct-to-Earth contingency beacon. The GNC compute stack (visual odometry → navigation EKF → hazard assessment → arc path planner → arc controller → mode manager/FDIR) is the software layer that actually closes the loop between all of these physical interfaces every drive arc.

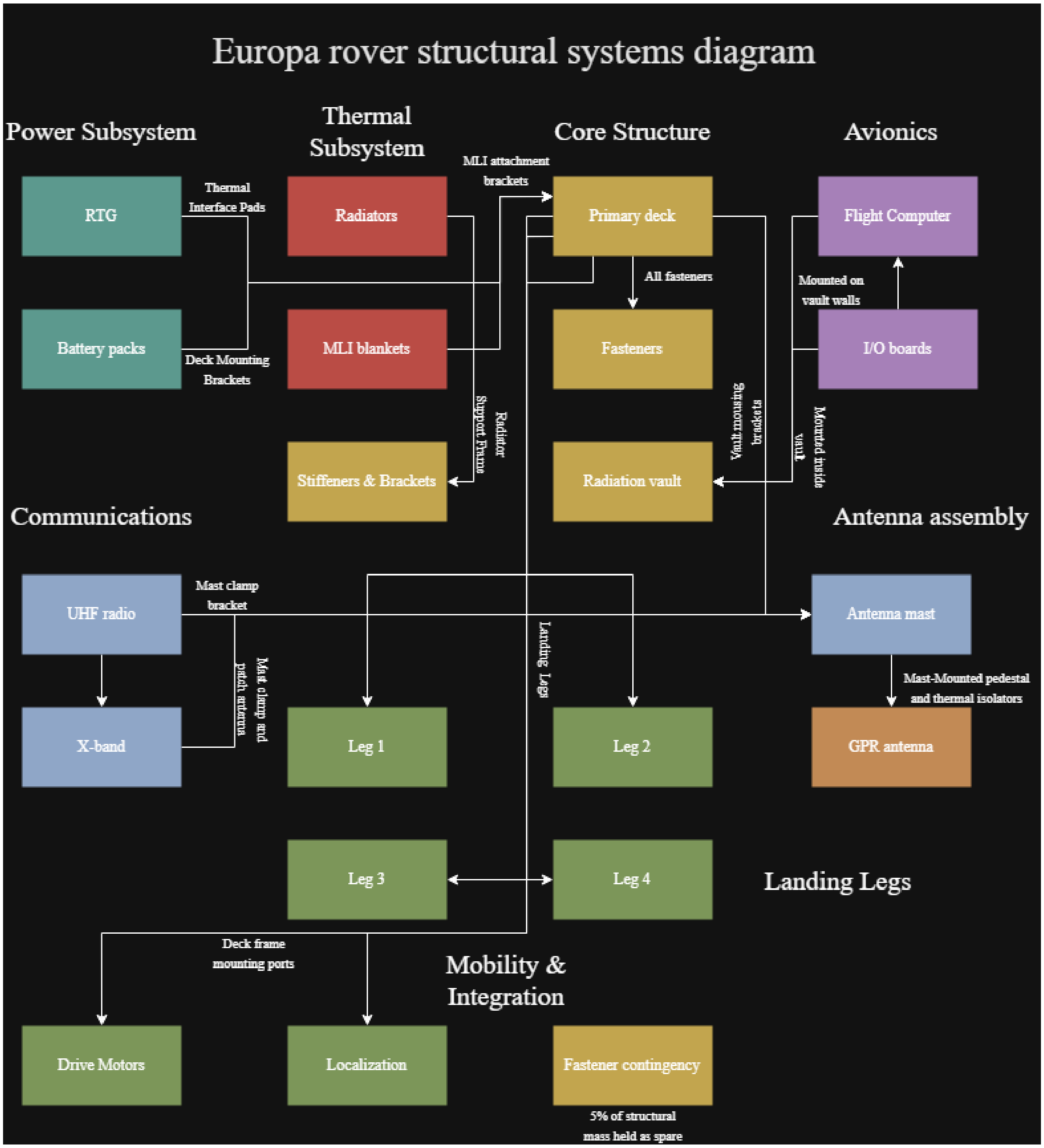

Figure 2: Rover systems diagram: subsystem-level interfaces across Power, Thermal, Core Structure, Avionics, Communications, Antenna Assembly, Landing Legs, and Mobility & Integration.

7.3 Structures and Mechanisms

7.3.1 Structural Requirements

- **REQ.03.02** — Antenna boresight within ±5° of nadir during sounding; rover halts traversal above 20° surface slope. (Structures, Science/Payload, Mobility)
- **REQ.07.01** — The rover chassis and landing leg structure shall accommodate vertical landing loads of 12g and lateral loads of 6g touchdown, with each leg sized to distribute peak contact

forces across four footpads without permanent deformation or buckling. (Structures, Mobility, GNC)

- **REQ.07.02** — Primary equipment deck, fasteners, and all structural material shall remain within specification (yield strength, fatigue life) under sustained 90 K cryogenic operation over the 30-sol mission lifetime. (Structures, Thermal, Power)
- **REQ.07.03** — The radiation shielding vault enclosing the flight computer and power distribution unit shall attenuate the Europa radiation environment to ≤ 1 krad/sol at the electronics level, maintaining structural stiffness and mass allocation within the structures budget. (Structures, Power, Electronics, CDH)

GAP — Needs Author Input

REQ.07.03's literal text allocates the radiation vault a 28.5 kg share of the structures budget, but the Mass Budget rollup (Table 11) totals the entire Structure subsystem at only 25.75 kg. Either the 28.5 kg figure in the requirement is stale, or the mass budget spreadsheet needs a dedicated vault-allocation line to reconcile against it – resolve this discrepancy before final submission. Separately, **REQ.05.02** (CDH latch-up recovery) is referenced by both REQ.07.03's validation plan ("latchup detection/recover per REQ.05.02") and REQ.08.03's reasoning field, consistently pointing to an autonomous power-cycling recovery within roughly 330 s of a radiation-induced latch-up, but its literal "shall" statement has still not been transcribed – pull it from the master spreadsheet.

7.3.2 Structural Sketch

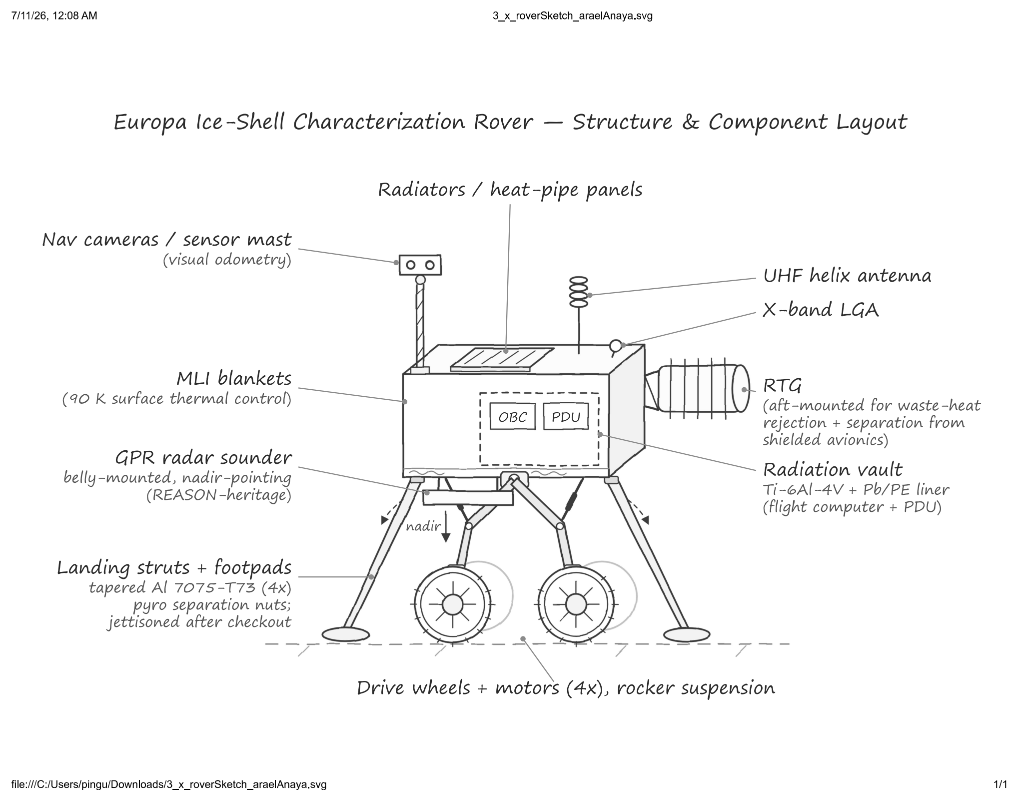

Figure 3: Annotated structural sketch of the rover chassis, deck, radiation vault, antenna mast, and landing-leg arrangement.

7.3.3 Mass Budget

Table 10 reproduces the mass budget in full. Dry mass totals 117.8 kg against a 147.25 kg contingency cap (25% allowable mass growth, 29.45 kg).

Subsystem	Component	Qty	Mass (kg)	Notes
Structure	Primary equipment decks/plates	2	5.24	Parametric sizing for avionics + GPR antenna footprint
Structure	Landing legs (struts + footpads)	4	10.36	2.59 kg/leg, tolerates $< 12g$ vertical / $6g$ lateral touchdown
Structure	Stiffeners/brackets/frames	1	1.88	Prevents deck deflection under landing loads
Structure	Fasteners & support hardware	1	1.42	Cryogenic-rated titanium M5/M6
Structure	Fastener contingency reserve	1	0.73	5% structural mass contingency
Structure	Antenna mast & mounting bracket	1	2.14	GPR pedestal, Al tube + thermal isolators for nadir pointing
Structure	Radiation shielding vault enclosure	2	2.08	Ti-6Al-4V; attenuates dose per REQ.01.02
Structure	Lead/polyethylene radiation liner	1	1.56	Selective shielding for flight computer
Structure	Thermal interface pads	2	0.34	Minimizes thermal short through vault wall at 90 K
Thermal	MLI attachment structure	1	1.44	Brackets/clips for MLI retention + radiator support
Thermal	Radiator support frame	2	1.52	Al tube frame, thermal decoupling
Thermal	MLI blankets	1	6.32	0.35 kg/m^2, 18.05 m^2 equivalent coverage
Thermal	Radiators / heat-pipe panels	2	9.58	4.79 kg/panel @ 6.0 kg/m^2, sized for 400–500 W rejection
Power	Battery packs (12-hr survival)	1	19.2	150 Wh/kg $\times$ 16 kg +20% margin
Power	PDU + high-current cabling	1	4.68	Heritage EPS scaling
Power	RTG	1	28.0	Main power source
Power	Heater	1	2.86	Heating unit
Comms	UHF radio + cabling	2	1.83	Heritage UHF radios, 1.5–2 kg each
Comms	X-band transponder	1	3.02	DSN-compatible SDST heritage
Comms	Antennas (UHF helix / X-band LGA)	1	0.86	Short RF cabling
Avionics	Flight computer	1	2.54	OBC board + enclosure
Avionics	I/O / storage / interface boards	1	1.52	Logger + ADCs, includes enclosure
Payload	MEDA-class environment suite	1	5.5	Perseverance MEDA heritage, ~ 17 W op. power
Software	Flight / GNC / FDIR	0	0	No physical mass; impacts OBC/EPS sizing only
Harness & Integration	System harness / connectors / strain relief	1	3.18	2–3% of dry mass
			Dry Mass Total	**117.80 kg**
			Allowable Mass Growth (25%)	**29.45 kg**
			Mass Cap Limit	**147.25 kg**

Table 10: Rover mass budget by subsystem and component.

Subsystem	Total Mass (kg)	% Dry Mass
Structure	25.75	21.86
Thermal	18.86	16.01
Power	54.74	46.47
Comms	5.71	4.85
Avionics/OBC	4.06	3.45
Payload	5.50	4.67
Harness and Integration	3.18	2.70
Software	0.00	0.00
Total	**117.80**	**100.00**

Table 11: Mass budget rollup by subsystem.

GAP — Needs Author Input

The mobility solution (Section 7.7) adds an estimated 17.1 kg (wheels, rocker suspension, drive motors, mast pan/tilt actuator) and 3.5 W that are *not yet* reflected as explicit line items in Table 10/11 above – the mobility document itself flags this directly and notes the budget was already sitting near 136.9 kg against the 147.25 kg cap before that addition. Reconcile the mass budget spreadsheet to add a dedicated "Mobility" subsystem row (or fold the drive train into the existing Structure/Power rows) before this table is considered final, and re-check margin against the cap.

7.3.4 Structures Trade Study

Study: Analysis of Landing Leg Structure. Options were scored against six pairwise-weighted criteria (Load Bearing Capability 0.25, Mass per Leg 0.20, Safety Margin at Yield 0.25, Thermal Conductivity 0.18, Manufacturability & TRL 0.25, Cryogenic Ductility 0.05).

Criterion (weight)	Tapered Strut Al 7075-T73	Cross-Braced Tube + Diagonals	Ti-6Al-4V F
Load Bearing (0.25)	1.25	1.25	
Mass per Leg (0.20)	0.80	0.60	
Safety Margin at Yield (0.25)	1.25	1.25	
Thermal Conductivity (0.18)	0.54	0.36	
Manufacturability & TRL (0.25)	1.25	0.75	
Cryogenic Ductility (0.05)	0.20	0.25	
Total Score	**5.29**	4.46	

Table 12: Landing leg structure trade study.

Winning option: Baseline tapered-strut Al 7075-T73 leg. It matches the top score on Load Bearing Capability and Safety Margin at Yield (both 5/5) while separating itself on Manufacturability & TRL (5/5, off-the-shelf/heritage design, simple machining), which is the largest single differentiator against the cross-braced and folding-hinge alternatives. This is consistent with the 2.59 kg/leg figure and material call-out already carried in the mass budget (Table 10) and the tapered-strut geometry drawn in Figure 3.

7.4 Electrical Power System

7.4.1 Power Requirements

- **REQ.05.03** — Per-sol energy draw across all subsystems shall not exceed 90% of the RTG's nominal daily energy output, reserving ≥ 10% margin for anomaly response, heater activations, and unplanned relay passes. (Power, CDH, Thermal, Mobility)

- **REQ.08.01** — The power system shall generate ≥ 110 We (BOL) via an MMRTG-class RTG and deliver ≥ 80 We continuous upon Europa arrival, providing ≥ 1.9 kWh usable energy per sol across the 30-sol surface mission. (Power, Thermal, System)

- **REQ.08.02** — Energy storage shall provide ≥ 2.4 kWh usable capacity to (1) load-level peak draws (drive motors, radar transmission, X-band downlink) beyond continuous RTG output, and (2) sustain night-mode survival heating and critical avionics for ≥ 12 hours, maintaining cell temperature above −20°C per REQ.05.01. (Power, Thermal)

- **REQ.03.03** — Dedicated peak power allocation reserved for radar transmission pulses; drive motors commanded off during each active sounding window. (Science/Payload, Power, Mobility)

- **REQ.08.03** — The power distribution unit shall autonomously detect and isolate a faulted load bus within 100 ms of overcurrent/undervoltage detection, maintain uninterrupted power to critical loads (flight computer, survival heaters, UHF radio), and log each fault event to non-volatile memory, without ground intervention. (Power, CDH, Electronics)

7.4.2 Power Budget & Profile

Table 13 gives the per-component power budget; the RTG delivers a constant 110 W generated power baseline against which every mode is evaluated. Seven operating modes were simulated across a 24-hour sol: Traverse Planning (1 hr), Traverse (4 hr), Radar Sounding (5 hr), Comms Relay Pass (1.5 hr), Map Generation & Consensus (4.5 hr), and Night Low-Power (8 hr), summing to the full 24-hour cycle.

Subsystem	**Component**	**Peak (W)**	**Avg (Wh/hr)**	**Duration (h)**
Payload	GPR Radar Sounder	45	225.0	5.0
EPS	Battery Packs	0	0.0	24.0
EPS	PDU (PCDU)	12	288.0	24.0
Mobility	Drive Motors (4x)	84	336.0	4.0
GNC	Flight Computer (OBC)	18	432.0	24.0
GNC	I/O & Storage Boards	6	96.0	16.0
GNC	Nav Cameras	4	20.0	5.0
Thermal	Heaters	40	320.0	8.0
Thermal	Thermal Sensors	0.5	12.0	24.0
Comms	UHF Radio	35	210.0	6.0
Comms	X-band Transponder	45	67.5	1.5
Total			**2006.5 Wh/sol**	

Table 13: Power budget by component (peak draw and average energy consumption per sol).

Mode	Traverse	Sounding	Comms Relay	Map Gen.	Traverse Plan.	Warm-Up	Night
Avg. Power (W)	124.5	81.5	116.5	71.5	40.5	121.5	70.5

Table 14: Average power draw by operating mode.

The battery (Max Charge 2400 Wh, Min Safe Charge 480 Wh) never approaches its safety floor across the simulated 24-hour sol: lowest observed charge is 2342.1 Wh (only $\sim$ 58 Wh, or 2.4%, below full), average power usage is 93.92 W against 110 W generated, and highest charge/discharge rates are 69.5 W/14.5 W respectively. This is consistent with REQ.05.03's 90%-of-RTG-output cap with 10% reserve: $93.92/110 = 85.4\%$, inside the cap with margin to spare. Drive motors and the radar sounder are never scheduled concurrently in the mode matrix (REQ.03.03's stop-and-sound logic), and the flight computer, PDU, and thermal sensors stay powered continuously across every mode as the "always-on" core avionics load.

GAP — Needs Author Input

The Power Budget sheet's own annotations note its battery-pass timing rows ("passes Hawaii roughly every 21 minutes," "count Hawaii for a maximum of about 7 minutes") appear to be leftover boilerplate from a satellite ground-pass planning template that was not fully adapted to the Europa mission context. These notes do not affect the power totals above, but the underlying spreadsheet should be cleaned up before submission so it does not read as Earth-orbit-specific.

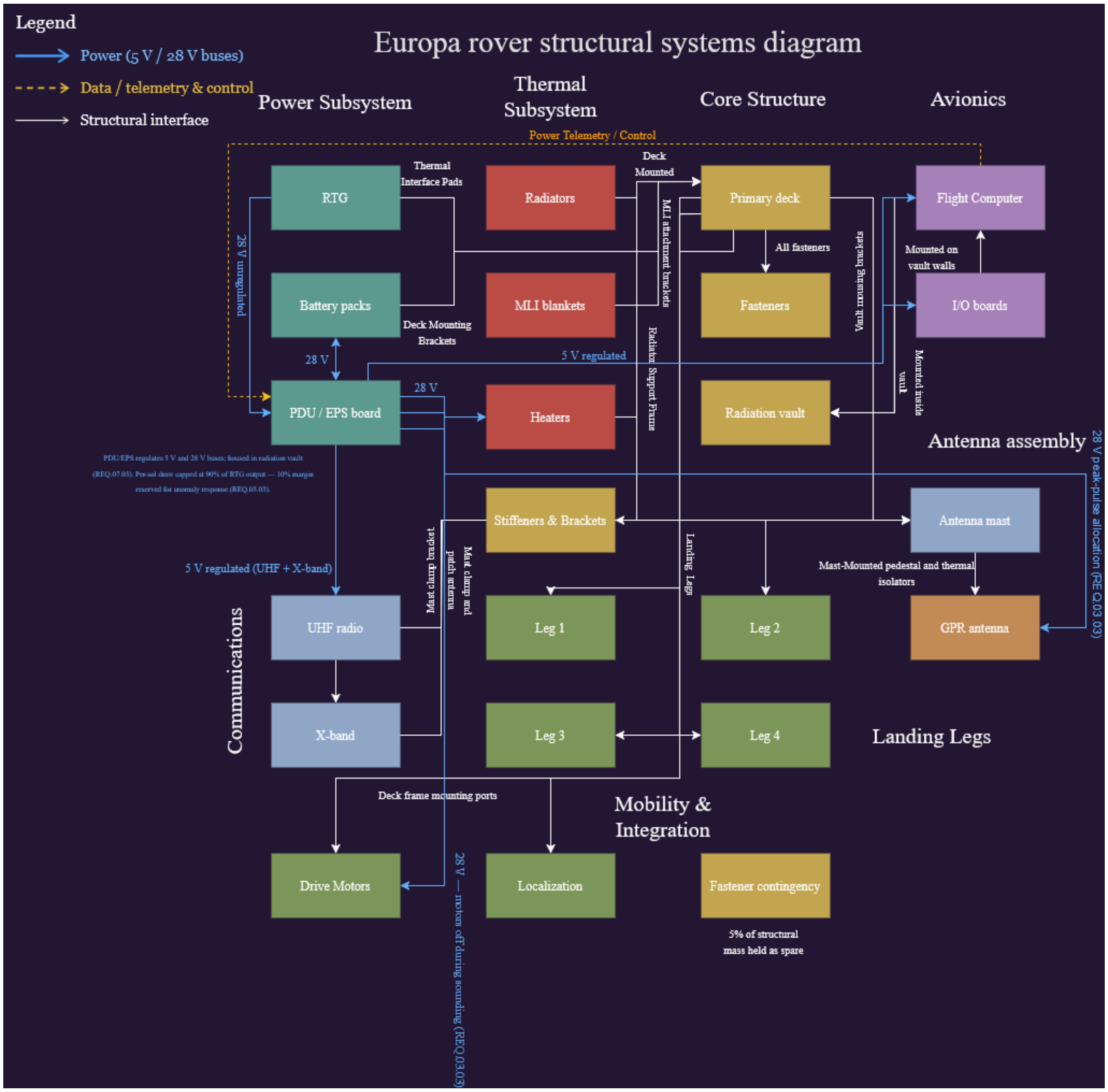

Figure 4: Power systems diagram: RTG and battery packs feeding the PDU/EPS board over regulated 28 V/5 V buses, with per-sol draw capped at 90% of RTG output (REQ.05.03) and peak-pulse allocation reserved for the radar sounder (REQ.03.03).

7.4.3 Power Trade Study

Study: Analysis of Power Source. Options were scored against five pairwise-weighted criteria (Mass 0.13, Continuous Power 0.28, Complexity/Operations 0.13, Radiation & Cryo Robustness 0.25, Availability/TRL 0.23).

Criterion (weight)	**Solar Arrays + Li-ion**	**RTG**	**H_2/O_2 Fuel Cells**
Mass (0.13)	0.25	0.38	0.13
Continuous Power (0.28)	0.28	1.38	0.55
Complexity/Operations (0.13)	0.25	0.63	0.25
Radiation & Cryo Robustness (0.25)	0.50	1.25	0.75
Availability/TRL (0.23)	0.68	1.13	0.68
Total Score	1.95	**4.75**	2.35

Table 15: Power source trade study.

Winning option: RTG (radioisotope thermoelectric generator). At 5.2 AU, solar irradiance has collapsed too far for arrays to deliver continuous power, and the RTG scores a perfect 5/5 on four of the five criteria (Continuous Power, Complexity/Operations, Radiation & Cryo Robustness, and Availability/TRL), losing only on raw mass (score 3/5, still the highest-weighted criterion it does not win). This selection is what sizes the 28 kg RTG line item in the mass budget (Table 10) and directly supports REQ.08.01's $\geq$ 110 We beginning-of-life requirement.

7.5 Thermal Management

7.5.1 Thermal Requirements

- **REQ.05.01** — Maintain avionics and battery cells within 0–40°C during active operations and above −20°C during night-mode low-power management, using RTG-waste-heat-powered active heater elements. (Thermal, Power, Structures)
- **REQ.09.01** — Reject up to 500 W of combined electronics dissipation and routed RTG waste heat through dedicated radiator panels, keeping the avionics vault below 40°C during peak daytime sounding/traverse operations. (Thermal, Electronics, Power, CDH)
- **REQ.09.02** — Preheat all drive-train actuators and deployment mechanisms above −40°C before any commanded motion, completing cold-start warm-up within 60 minutes of the 90 K surface baseline. (Thermal, Mobility, Structures)
- **REQ.09.03** — Combine MLI with routed RTG waste heat to bound the equipment-enclosure heat leak such that supplemental electric survival heating does not exceed 50 W and stays within the REQ.05.03 per-sol energy budget. (Thermal, Power, Structures)

7.5.2 Heat Load Budget

Table 16 lists per-component power usage, efficiency, and resulting heat generation. Aggregate power usage across all listed components is 289.5 W at a blended efficiency of 0.942, for a total heat generation of 16.91 W (this is the electronics dissipation term; it excludes the RTG's own ~ 2 kW continuous thermal rejection, which is routed separately as the primary survival heat source per REQ.09.03).

Board	Component	Power (W)	Eff.	Heat Gen. (W)
Payload	GPR Radar Sounder (REASON-heritage)	45	0.95	2.25
EPS	Battery Packs	0	1.00	0.00
EPS	PDU (PCDU)	12	0.93	0.84
Mobility	Drive Motors (4x)	84	0.90	8.40
GNC	Flight Computer (OBC)	18	0.95	0.90
GNC	I/O & Storage Boards	6	0.95	0.30
GNC	Nav Cameras (stereo VO)	4	0.95	0.20
Thermal	Heaters	40	1.00	0.00
Thermal	Thermal Sensors	0.5	0.95	0.025
Comms	UHF Radio (Electra-Lite heritage)	35	0.95	1.75
Comms	X-band Transponder	45	0.95	2.25
Total		**289.5**	**0.942**	**16.91**

Table 16: Heat load budget by component.

In addition to the equipment heat load above, a point-mass thermal equilibrium analysis was performed for the avionics enclosure (area 0.12 m^2, $\alpha = 0.20$, $\varepsilon = 0.85$). The hot case ($Q_{int} = 20$ W, daytime operations) equilibrates at $T_{hot} \approx 248$ K (-25°C); the cold case ($Q_{int} = 5$ W, night survival) equilibrates at $T_{cold} \approx 175$ K (-98°C) – a ~ 73 K swing. Unlike a Mars-class mission, the problem on Europa is not rejecting heat but retaining it: solar irradiance at 5.2 AU collapses to ~ 50 W/m^2, so internal dissipation dominates the energy balance, and even the operating case falls below the ~ 253 K (-20°C) lower bound for Li-ion battery charging. The survival case sits far outside any electronics or battery limit and would freeze the system without intervention – which is exactly why MLI, routed RTG waste heat, and controlled survival heaters (REQ.09.03) are required rather than optional.

7.5.3 Thermal Trade Study

Study: Analysis of Night Survival Thermal Architecture. Options were scored 1–5 against five pairwise-weighted criteria (Mass 0.13, Power Demand 0.25, Complexity/Integration 0.13, Radiation Durability 0.25, Reliability/TRL 0.25).

Criterion (weight)	MLI + Heaters	Aerogel + Heaters	PCM + Heat Pipes
Mass (0.13)	0.52	0.52	0.26
Power Demand (0.25)	1.25	0.75	1.00
Complexity/Integration (0.13)	0.52	0.39	0.26
Radiation Durability (0.25)	1.25	0.75	0.75
Reliability/TRL (0.25)	1.25	0.75	0.50
Total Score	**4.79**	3.16	2.77

Table 17: Night-survival thermal architecture trade study.

Winning option: MLI and heaters (4.79 vs. 3.16 for aerogel + heaters and 2.77 for PCM with heat pipes). RTG waste heat (~ 1.9 kW thermal) serves as the primary survival heat source;

MLI limits radiative loss so that supplemental electric survival heating stays at or below 50 W (REQ.09.03), keeping night thermal draw inside the per-sol energy budget (REQ.05.03).

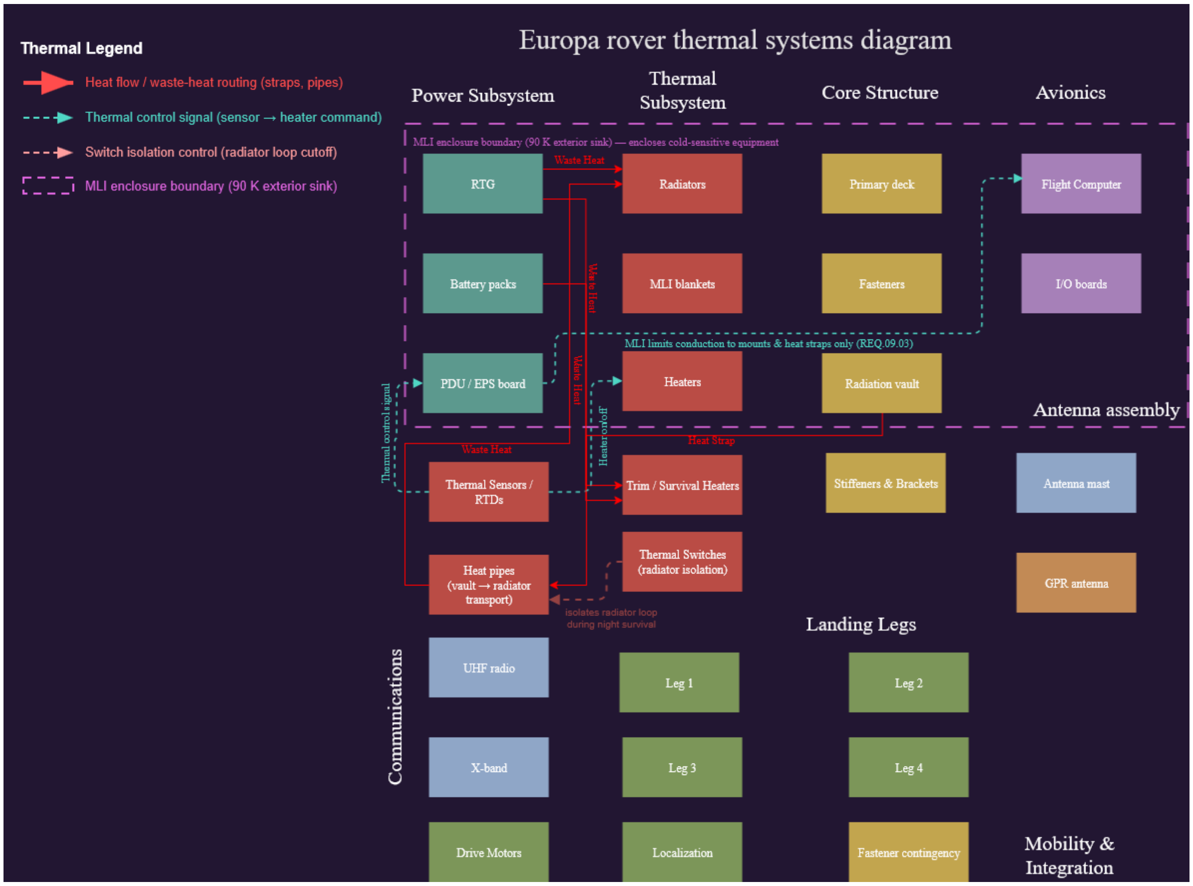

Figure 5: Thermal systems diagram: RTG/heater waste-heat routing to radiators, MLI enclosure boundary, and radiator-loop isolation switches for night survival.

7.6 Communications

7.6.1 Communications Requirements

- **REQ.01.03** — Maintain ≥ 2 active relay nodes at all times during surface operations. (System, COMMs)
- **REQ.04.01** — Inter-agent mesh network shall achieve terrain-map consensus within ≤ 10 minutes under $\leq 30\%$ packet loss on any single link. (COMMs, CDH)
- **REQ.10.01** — Deliver ≥ 55 MB/Earth-day compressed data at ≥ 128 kbps return / ≥ 8 kbps forward. (COMMs, CDH, Science/Payload, Power)
- **REQ.10.02** — Obtain ≥ 2 usable relay contacts/sol totaling ≥ 90 min above a 10° elevation mask; end-to-end latency ≤ 26 hours. (COMMs, CDH, Science/Payload, GNC)
- **REQ.10.03** — Close the proximity link with ≥ 3 dB margin at the worst-case 588 km slant range; retain an X-band DTE beacon ≥ 5 bps as a relay-loss fallback. (COMMs, Power, Structures, CDH)

7.6.2 Comm Architecture & Assumptions

The rover's next communication node is the nearer of the two orbital relay agents; the mission does not route science data through a direct link to Earth in nominal operations. The link uses a UHF proximity band on the CCSDS Proximity-1 return/forward pair: 401.6 MHz return, 437.1 MHz forward, the same split flown by Electra and Electra-Lite on the Mars Relay Network. Desired margin is 3 dB above the E_b/N_0 required for a post-decoding bit error rate of 10^{-6} (CCSDS turbo coding, rate 1/2, threshold near 1.5 dB, so the link must clear 4.5 dB total) – a conservative cushion given that the relay's exact orbit geometry, the rover's helix gain pattern, and Jupiter's synchrotron noise contribution are all still modeled rather than measured. Worst-case geometry assumes both relays fly a 200 km circular Europa orbit; with a 10° elevation mask, the slant range at the edge of a pass is 588 km, and every link budget term uses that worst case.

Rover transmit power is 5 W RF into a 5.0 dBi quadrifilar helix (near-hemispherical pattern, so the rover can close a pass without pointing at anything); the relay carries 8 W RF into a 7.0 dBi patch array, since its power budget has more slack. System noise temperature is 3000 K at the relay receiver and 4000 K at the rover receiver – the number that most separates a Europa proximity link from a Mars one, since at 400 MHz the receiver picks up galactic background noise on top of Jupiter's synchrotron emission. The return link runs 128 kbps (sized to clear 55 MB in the 90-minute relay window) and the forward link runs 8 kbps (commands only). Figure 6 shows the full data-flow architecture from the payload through CDH, the modem/transceiver chain, the RF front end, and out to the relay agents and, ultimately, the DSN.

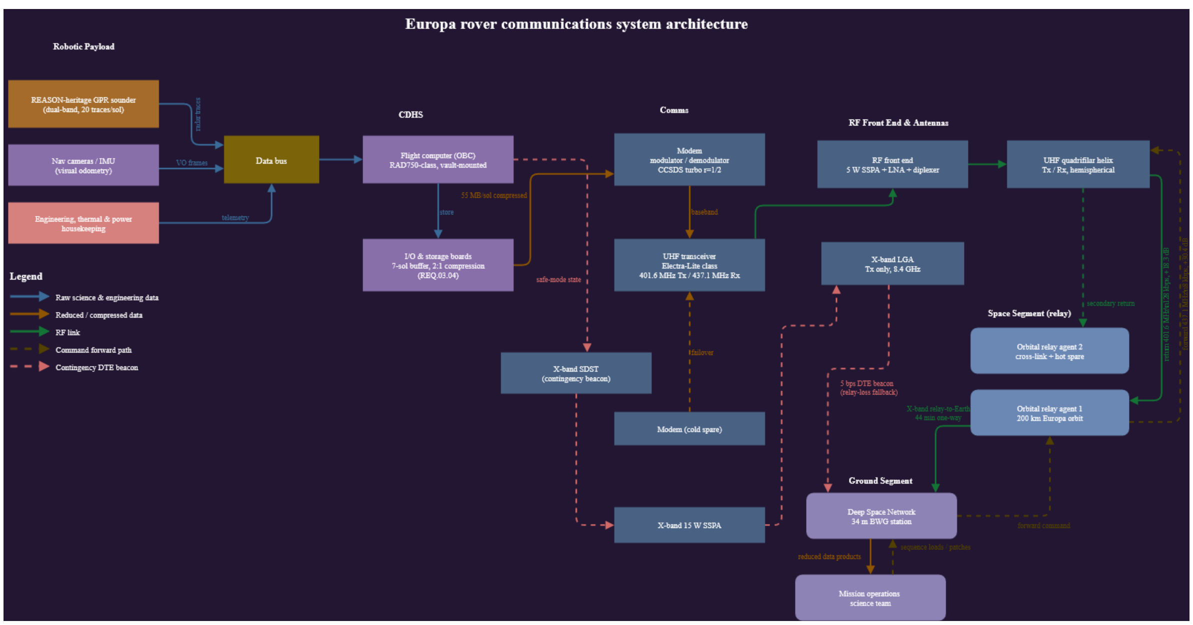

Figure 6: Communications system architecture: data flow from payload/nav/health sources through CDH, comms, and the RF front end to the orbital relay agents and DSN ground segment.

7.6.3 Partial Link Budget

Both directions close with wide margin: 18.3 dB on the return (uplink, rover→relay) and 30.4 dB on the forward (downlink, relay→rover), against a 4.5 dB requirement. Table 18 breaks the return link down term by term.

Term	Value (dB)
P_t (5 W)	+6.99
G_t (rover helix)	+5.00
G_r (relay patch array)	+7.00
L_a (polarization + plasma)	−0.51
L_l (line loss)	−1.49
λ^2	−2.54
$(4\pi R)^2$ (space loss @ 588 km)	−137.37
k	+228.60
T_s (3000 K)	−34.77
R_b (128 kbps)	−51.07
E_b/N_0	**+19.84**
Margin over 1.5 dB requirement	**+18.3**

Table 18: Return-link (rover-to-relay) budget, term by term.

As a direct-to-Earth sanity check: the rover's X-band low-gain antenna against a DSN 34 m station closes at roughly 6.9 bps at opposition (4.2 AU) and 3.2 bps at conjunction (6.2 AU) while holding the same 4.5 dB margin – enough for a state-of-health beacon, but a single 55 MB sol of science data would take over two years to clear at that rate. This is why the relay architecture is foundational rather than a convenience (REQ.10.03's 5 bps fallback floor is set directly from this analysis).

7.6.4 Communications Subsystem Diagram

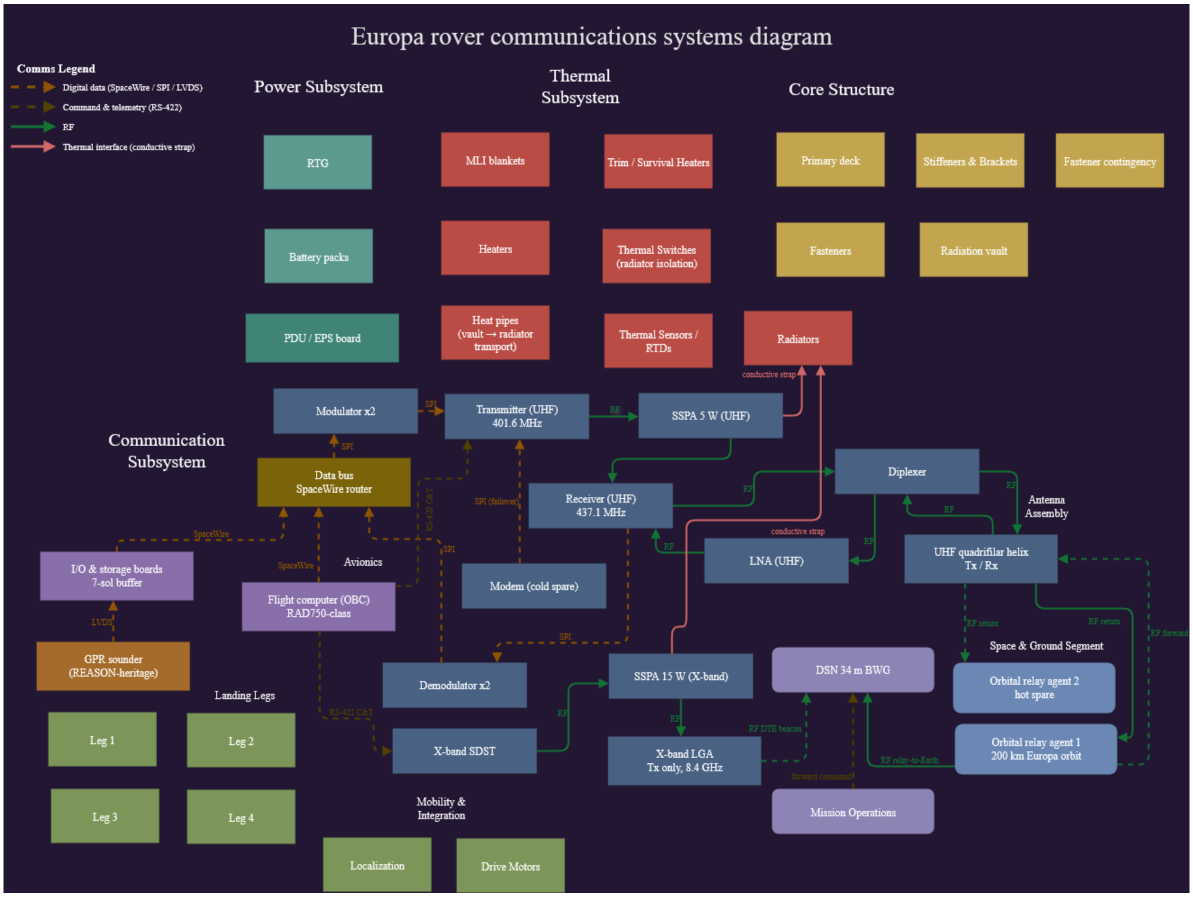

Figure 7: Communications systems diagram: modulator/transceiver chain, RF front end, and antenna assembly, with digital-data, command/telemetry, RF, and thermal-interface signal paths distinguished.

7.6.5 Communications Trade Study

Study: Analysis of Rover-to-Relay Proximity Link Configuration. Options were scored against five pairwise-weighted criteria (Mass 0.13, Power Demand 0.25, Data Rate Capability 0.25, Pointing & Ops Complexity 0.13, Reliability/TRL 0.25).

Criterion (weight)	UHF Helix	S-band Patch	X-band Gimballed HGA
Mass (0.13)	0.65	0.52	0.26
Power Demand (0.25)	1.00	0.75	0.50
Data Rate Capability (0.25)	1.00	1.25	1.25
Pointing & Ops Complexity (0.13)	0.65	0.52	0.13
Reliability/TRL (0.25)	1.25	0.75	0.50
Total Score	**4.55**	3.79	2.64

Table 19: Rover-to-relay proximity link configuration trade study.

Winning option: UHF quadrifilar helix (401.6/437.1 MHz). The X-band option has the raw throughput but fails the other criteria: a two-axis gimbal is a moving mechanism that has to survive 90 K and 100 krad, and it needs closed-loop tracking of a relay the rover cannot see coming. S-band buys data rate the mission does not need, since the UHF return already closes at 128 kbps with 18.3 dB of margin and is bounded by the transceiver's 2048 kbps ceiling rather than by RF. The helix is already in the mass budget at 0.86 kg and carries TRL 9 through Electra and Electra-Lite heritage on exactly this kind of proximity link.

7.7 GNC and Mobility

7.7.1 GNC Requirements

- **REQ.11.01** — Determine attitude to $\pm 0.5°$ (3σ) roll/pitch and heading to $\pm 0.25°$ (3σ) at every sounding stop, propagating inertially between arcs ≤ 5 m and taking an absolute heading fix from fine sun sensors at each arc end. (GNC, Mobility, Science/Payload, CDH)
- **REQ.11.02** — Assess terrain ≥ 8 m ahead via stereo nav cameras before each drive arc; autonomously reject arcs with a step > 0.20 m, slope $> 20°$, or roughness > 0.7; stop within 0.25 m of the assessed safe limit. (GNC, Mobility, Science/Payload, CDH)
- **REQ.11.03** — Follow the planned path with cross-track deviation ≤ 0.5 m over a 50 m segment and stop within ± 0.5 m of each commanded waypoint, using differential drive with wheel slip estimated from the odometry/visual-odometry residual. (GNC, Mobility, Science/Payload)
- **REQ.11.04** — Halt and enter a navigation hold whenever 1σ position uncertainty exceeds 0.5 m; re-initialize via two-way ranging to ≥ 2 relay agents plus local/orbital terrain correlation; do not resume until uncertainty is restored below 0.5 m. (GNC, COMMs, CDH, Mobility)

7.7.2 Mobility Solution Sketch

Configuration: Four-Wheel Differential Drive on a Single-Pivot Rocker. Four independently driven wheels ride on a single-pivot rocker, with no steering actuators. The chassis is the 0.90 m × 0.60 m equipment deck already carried in the mass budget, held 0.35 m above the surface so the belly-mounted sounder keeps its standoff. Wheelbase is 0.70 m and track is 0.80 m; wheels are 0.32 m in diameter, 0.20 m wide, with a rigid aluminum rim carrying 24 hardened cleats. Commanded speed is 2 cm/s, driven in arcs no longer than 5 m.

Wheels were chosen over hopping or legged locomotion because the mission needs a spatially continuous subsurface profile: REQ.03.01 calls for a radar trace every 100 m or better along a single transect, and every landing event in a hop or step creates a gap where the localization solution has to start over. A hop that lands wrong on a 90 K ice block can end the mission on the spot, with no window for Earth to intervene given the 44-minute one-way delay; a wheeled rocker instead degrades slowly, bearing by bearing, over many sols – a failure mode better suited to a mission with no recovery crew.

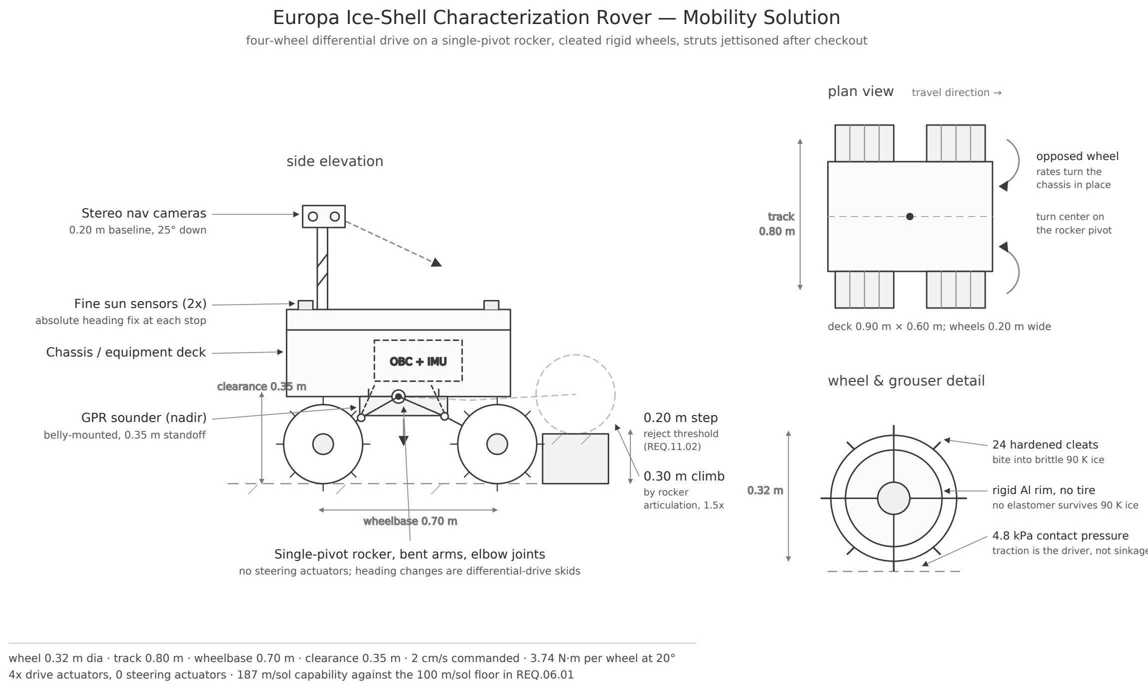

Figure 8: Mobility solution: side elevation, plan view, and wheel detail.

Steering is handled entirely by differential drive: opposing wheel rates rotate the chassis about the rocker pivot, so every heading change is a deliberate skid across the ice. Skid steering slips badly on hard ice ($\mu \approx 0.5$), so wheel odometry is dropped as a position source and used instead as a slip observable – the residual between encoder counts and stereo visual odometry becomes the slip estimate, the same approach the MER rovers used on Martian sand [14]. Heading is re-anchored with an absolute sun-sensor fix at the end of every 5 m arc so error never integrates across a full traverse segment.

7.7.3 Localization and Pointing Budget

Table 20 closes the 50 m segment localization error budget: root-sum-square of 0.46 m against the 0.5 m limit in REQ.02.01, an 8% margin.

Error source over a 50 m segment	Contribution (m)
Visual odometry scale and feature residual	0.30
Heading knowledge (0.271°)	0.24
Wheel slip on cleated water ice	0.20
Inter-agent range residual	0.15
Root-sum-square	**0.46**
REQ.02.01 limit	0.50

Table 20: Localization error budget, one 50 m traverse segment.

In addition to this position-error rollup, the source pointing budget defines per-component alignment requirements rather than a single combined pointing RSS: the GPR antenna boresight must hold ±5° of nadir (±0.5° stop-to-stop repeatability, REQ.03.02); the fine sun sensor pair resolves the Sun vector to ±0.1°, which folds into a ±0.25° heading solution after ephemeris and mounting alignment error (REQ.11.01); and the stereo nav camera boresight sits 25° below the local horizontal with a 0.20 m step-detection capability at 8 m range (REQ.11.02).

7.7.4 GNC Trade Study

Study: Analysis of the absolute heading reference for surface navigation. Options were scored against five pairwise-weighted criteria (Mass 0.12, Power Demand 0.23, Complexity/Cost 0.12, Accuracy 0.30, Environmental Tolerance 0.23).

Criterion (weight)	Fine Sun Sensor Pair	COTS Star Tracker	FOG Gyrocompassing
Mass (0.12)	0.60	0.48	0.24
Power Demand (0.23)	1.15	0.69	0.46
Complexity/Cost (0.12)	0.48	0.24	0.12
Accuracy (0.30)	1.20	1.50	0.90
Environmental Tolerance (0.23)	1.15	0.46	0.46
Total Score	**4.58**	3.37	2.18

Table 21: Absolute heading reference trade study.

Winning option: Fine sun sensor pair with onboard ephemeris. The star tracker wins on raw accuracy and loses everywhere else: its detector degrades measurably under the Jovian dose, it needs a baffle against Jupiter (which subtends $\sim 11.8°$ from the surface and never sets on the sub-Jovian hemisphere), and it costs more than the rest of the GNC sensor suite combined. Gyrocompassing is the interesting failure: a 0.01°/h FOG can in principle find north off Europa's 4.22°/h rotation, but that signal is only 28% of what the same instrument would see on Earth, and the fiber coil and light source are not qualified at 90 K or 100 krad. The sun sensor is the cheap answer: at 5.2 AU the Sun delivers only 50 W/m^2, but its angular diameter collapses to 0.103°, so a photodiode sensor is looking at a sharper point source than it would at Earth. Two of them cost 0.12 kg and under 1 W, and Cassini flew the same class of sensor further out than Jupiter.

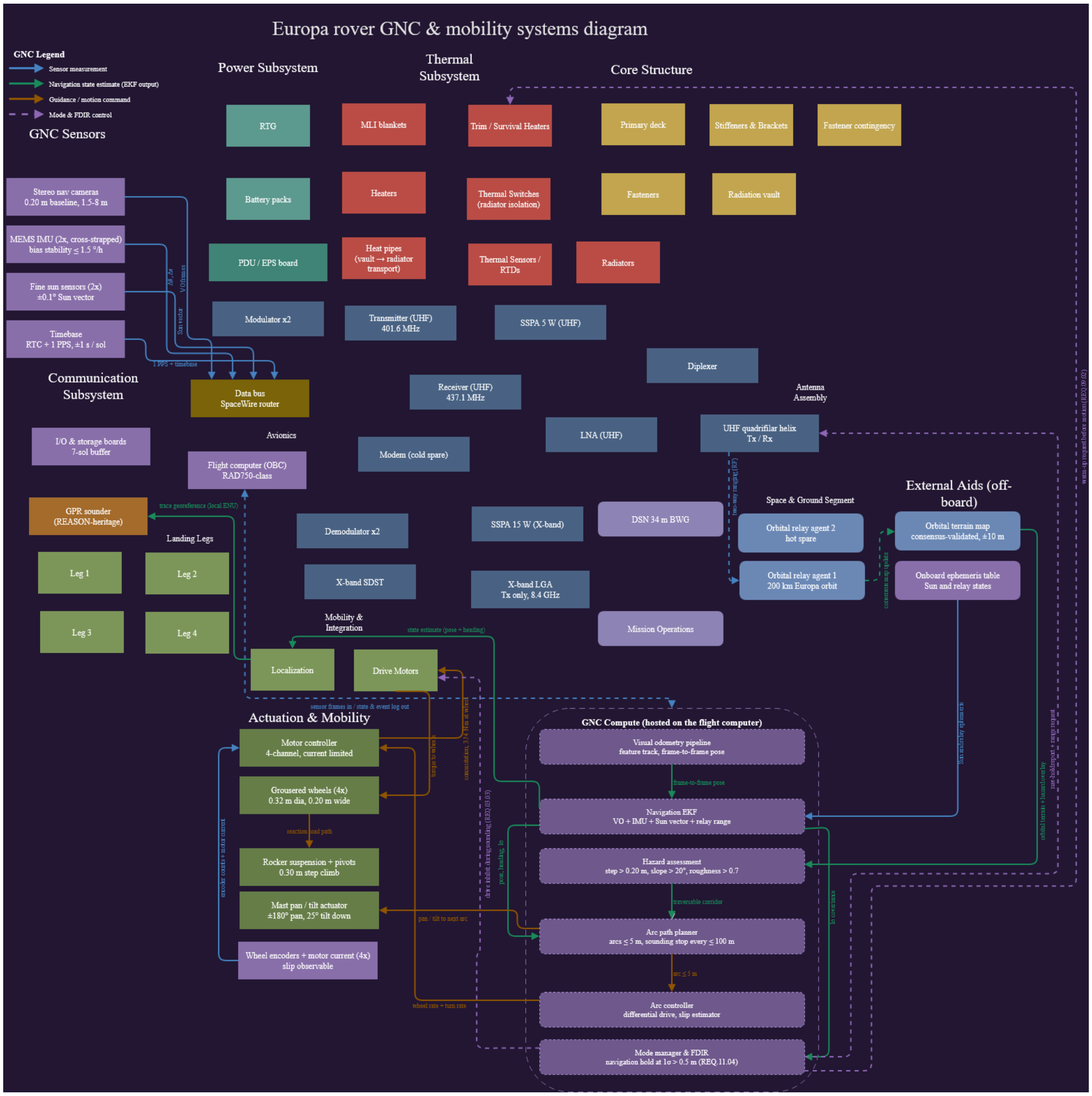

Figure 9: GNC and mobility systems diagram: sensor suite, GNC compute stack (visual odometry → navigation EKF → hazard assessment → arc planner → arc controller → mode manager/FDIR), and actuation/mobility hardware.

7.8 Command and Data Handling (CDH)

GAP — Needs Author Input

No dedicated CDH design assignment exists yet in this repository (there is no "3.7 Command and Data Handling Design" folder analogous to 3.2–3.6). Everything in this subsection is *reconstructed* from CDH-adjacent requirements and mentions scattered across the Comms, GNC, and payload deliverables – it is a placeholder sized to show the rubric's required elements exist in outline, not a completed CDH design. Before submission this subsection needs its own requirements table, a real data budget/storage-strategy write-up, a CDH systems diagram (matching the house style of Figures 4, 5, and 7), and a CDH trade study (e.g., flight computer or storage-media options).

7.8.1 CDH Requirements (reconstructed)

- **REQ.03.04** — Retain ≥ 7 sols of uncompressed radar trace data onboard; apply lossless compression at $\geq 2{:}1$ before inter-agent relay transmission. (CDH, Science/Payload, COMMs)
- **REQ.04.01** — Support mesh-network consensus on shared terrain maps within ≤ 10 minutes under degraded ($\leq 30\%$ packet loss) link conditions. (CDH, COMMs)
- **REQ.05.02** — *(Referenced but not verbatim-confirmed; understood to require autonomous recovery from a radiation-induced latch-up event within roughly* 330 *s via power-cycling, without ground intervention. Confirm literal text against the master spreadsheet.)* (CDH, Electronics)

7.8.2 Data Budget and Storage Strategy (reconstructed)

The flight computer is a RAD750-class OBC, radiation-vault-mounted, with dedicated I/O and storage boards sized to buffer 7 sols of uncompressed radar trace data (REQ.03.04). At approximately 40 traces per sol across the four data-product sheets, raw science data accumulates faster than the non-deterministic Earth relay link can clear, which is the entire justification for the 7-sol buffer. Lossless compression at $\geq 2{:}1$ is applied before inter-agent relay transmission, and the resulting compressed volume (≥ 55 MB/day, REQ.10.01) is what sizes the 128 kbps return-link data rate in Section 7.6. A safe-mode command path exists directly from the flight computer to the transceiver chain (bypassing the primary science-data path) so that a spacecraft-emergency state can always reach the relay network regardless of what state the science-data pipeline is in.

8 References

References

[1] National Academies of Sciences, Engineering, and Medicine, *Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023–2032*, The National Academies Press, Washington, D.C., 2022. Available: https://www.nationalacademies.org/read/26522.

[2] NASA Space Technology Mission Directorate, "Civil Space Technology Shortfall Rankings," NASA, Washington, D.C., Jul. 2024. Available: https://www.nasa.gov/wp-content/uploads/2024/07/civil-space-shortfall-ranking-july-2024.pdf.

[3] NASA, "FY 2026–2027 SBIR/STTR Program Solicitation, Appendix 26B-I: SBIR Topics," NASA Small Business Innovation Research Program, 2025. Topic COMNAV.1.S26B.

[4] NASA Jet Propulsion Laboratory, "Europa Clipper," JPL, Pasadena, CA. Available: https://www.jpl.nasa.gov/missions/europa-clipper/.

[5] Wikipedia contributors, "Europa Clipper," *Wikipedia, The Free Encyclopedia*. Available: https://en.wikipedia.org/wiki/Europa_Clipper.

[6] University of Texas at Austin Jackson School of Geosciences, "Scientific Instrument Developed by UTIG Scientists Headed to Europa," UT Austin, Austin, TX, Apr. 2025.

[7] NASA, "Radar for Europa Assessment and Sounding: Ocean to Near-surface (REASON)," Europa Clipper Mission. Available: https://europa.nasa.gov/spacecraft/instruments/reason/.

[8] Jet Propulsion Laboratory, "DSN Telecommunications Link Design Handbook," 810-005, Rev. E, NASA/JPL, Pasadena, CA. Available: https://deepspace.jpl.nasa.gov/dsndocs/810-005/.

[9] Consultative Committee for Space Data Systems, "Proximity-1 Space Link Protocol — Physical Layer," CCSDS 211.1-B-4, 2013.

[10] Consultative Committee for Space Data Systems, "TM Synchronization and Channel Coding," CCSDS 131.0-B-4, 2017.

[11] C. D. Edwards *et al.*, "The Electra Proximity Link Payload for Mars Relay Telecommunications and Navigation," Jet Propulsion Laboratory, Pasadena, CA.

[12] NASA, "Deep Space Network," NASA Space Communications and Navigation. Available: https://www.nasa.gov/communicating-with-missions/dsn/.

[13] A.-M. Kietzig, S. G. Hatzikiriakos, and P. Englezos, "Physics of Ice Friction," *Journal of Applied Physics*, vol. 107, no. 8, 081101, 2010.

[14] M. Maimone, Y. Cheng, and L. Matthies, "Two Years of Visual Odometry on the Mars Exploration Rovers," *Journal of Field Robotics*, vol. 24, no. 3, pp. 169–186, 2007.

[15] NASA Jet Propulsion Laboratory, "Mars Science Laboratory: Mobility System," JPL, Pasadena, CA. Available: https://mars.nasa.gov/msl/spacecraft/rover/wheels/.

[16] NASA Science, "Perseverance Rover Components," NASA Science Mission Directorate, Jul. 22, 2025. Available: https://science.nasa.gov/mission/mars-2020-perseverance/rover-components/.

Appendices

A Extended Trade Study Tables

The Structures (Table 12), Power (Table 15), Thermal (Table 17), Communications (Table 19), and GNC (Table 21) trade studies in Section 7 already reproduce the full pairwise-weighted decision matrices used in the source spreadsheets. Each uses the same methodology: criteria weighted by pairwise comparison in 0.25 increments (summing to 1 per pair, normalized to sum to unity across all criteria), and options scored 1–5 against a per-criterion rubric before weighting. Only the Structures and CDH scoring rubrics' full 1–5 band definitions are omitted here for length; they are reproduced in the source Decision Matrix / Trade Study PDFs in each subsystem's folder.

B Full Requirements Spreadsheet

Table 22 consolidates every requirement identified across the source assignments (REQ.01.01 through REQ.11.04), with owning subsystem(s). Full justification, verification, and validation text for each requirement is maintained in the master spreadsheet (*3.6 GNC and Mobility Design/Space Robotics Mission Sheet.xlsx*) and is not fully duplicated here for length; representative full-text entries appear in Sections 3.3 and 7.

ID	Statement (condensed)	Subsystem(s)
REQ.01.01	Full autonomy for entire surface mission; no real-time Earth uplink.	System
REQ.01.02	Survive $\geq$ 100 krad TID over mission lifetime.	System, Electronics, Power
REQ.01.03	$\geq$ 2 active relay nodes at all times during surface ops.	System, COMMs
REQ.02.01	Localization $\pm$0.5 m over any 50 m segment, no GPS.	Mobility, GNC
REQ.03.01	GPR resolves ice thickness $\pm$10%, 1–30 km depth, $\geq$ 100 m resolution.	Science/Payload, Power
REQ.03.02	Antenna boresight $\pm 5°$ nadir; halt above $20°$ slope.	Science/Payload, Structures, Mobility
REQ.03.03	Peak power reserved for radar pulses; drive shutdown during sounding.	Science/Payload, Power, Mobility
REQ.03.04	$\geq$ 7-sol onboard storage; $\geq$ 2:1 lossless compression before relay.	Science/Payload, COMMs, CDH
REQ.04.01	Mesh consensus $\leq$ 10 min under $\leq$ 30% packet loss.	COMMs, CDH
REQ.05.01	Avionics/battery 0–40°C active, $>$ -20°C night mode.	Thermal, Power, Structures
REQ.05.02	*(latch-up recovery ∼330 s, autonomous power-cycle — confirm literal text)*	CDH, Electronics
REQ.05.03	Per-sol draw $\leq$ 90% RTG output; $\geq$ 10% margin reserve.	Power, CDH, Thermal, Mobility

ID	Statement (condensed)	Subsystem(s)
REQ.06.01	$\geq$ 100 m/sol autonomous traverse; sounding stop $\leq$ 100 m intervals.	Mobility
REQ.06.02	*(site flagged when ice $<$ 5 km & slope $<$ 20° & roughness $<$ 0.7 — reconstructed from data-product doc; confirm literal text)*	Science/Payload, Mobility, GNC
REQ.06.03	Daily summary packet autonomously sent within 2 hr of sol's radar window.	COMMs, CDH, Science/Payload
REQ.07.01	Landing legs accommodate 12g vertical / 6g lateral touchdown loads across four footpads, no permanent deformation.	Structures, Mobility, GNC
REQ.07.02	Structural materials remain in-spec at 90 K over 30-sol lifetime.	Structures, Thermal, Power
REQ.07.03	Radiation vault attenuates dose to $\leq$ 1 krad/sol at electronics level within structures mass allocation.	Structures, Power, Electronics, CDH
REQ.08.01	MMRTG $\geq$ 110 We BOL, $\geq$ 80 We continuous, $\geq$ 1.9 kWh/sol.	Power, Thermal, System
REQ.08.02	Battery $\geq$ 2.4 kWh usable; $\geq$ 12 hr night survival; > -20°C.	Power, Thermal
REQ.08.03	PDU isolates a faulted load bus within 100 ms; critical loads uninterrupted; fault logged autonomously.	Power, CDH, Electronics
REQ.09.01	Reject $\leq$ 500 W; vault $<$ 40°C during peak ops.	Thermal, Electronics, Power, CDH
REQ.09.02	Preheat actuators > -40°C within 60 min of cold start.	Thermal, Mobility, Structures
REQ.09.03	MLI + RTG waste heat bound survival heaters $\leq$ 50 W.	Thermal, Power, Structures
REQ.10.01	$\geq$ 55 MB/day compressed; $\geq$ 128 kbps return / $\geq$ 8 kbps forward.	COMMs, CDH, Science/Payload, Power
REQ.10.02	$\geq$ 2 relay contacts/sol, $\geq$ 90 min total; latency $\leq$ 26 hr.	COMMs, CDH, Science/Payload, GNC
REQ.10.03	Link $\geq$ 3 dB margin @ 588 km worst case; X-band DTE beacon $\geq$ 5 bps fallback.	COMMs, Power, Structures, CDH
REQ.11.01	Attitude $\pm$0.5° roll/pitch, heading $\pm$0.25° per arc, sun-sensor fix every $\leq$ 5 m.	GNC, Mobility, Science/Payload, CDH
REQ.11.02	Hazard assessment $\geq$ 8 m lookahead; reject step $>$ 0.20 m / slope $>$ 20° / roughness $>$ 0.7.	GNC, Mobility, Science/Payload, CDH
REQ.11.03	Cross-track $\leq$ 0.5 m/50 m; waypoint $\pm$0.5 m; differential drive + slip estimate.	GNC, Mobility, Science/Payload
REQ.11.04	Navigation hold if 1σ $>$ 0.5 m; re-init via 2-relay ranging + map correlation.	GNC, COMMs, CDH, Mobility

ID	Statement (condensed)	Subsystem(s)

Table 22: Condensed master requirements list, REQ.01.01–REQ.11.04.

GAP — Needs Author Input

Entries above marked "confirm literal text" (REQ.05.02, REQ.06.02, REQ.07.01, REQ.07.03, REQ.08.03) were reconstructed from context in adjacent documents rather than transcribed directly from the master spreadsheet. Open *3.6 GNC and Mobility Design/Space Robotics Mission Sheet - Requirements.pdf* (the authoritative cumulative sheet), copy each row's literal statement, reasoning, and V&V text, and replace the reconstructed text here and in the relevant subsystem subsection of Section 7.

C Sample Datasets

GAP — Needs Author Input

Only the single illustrative Radar_Traces record reproduced in Table 7 (Section 6.2) is available at this time, and it was AI-generated to demonstrate the schema. Attach a short exported run (10–20 rows) from *Data Product Space Robotics.xlsx* covering at least the Radar_Traces and Daily Summary sheets before submission.

D Tools Used

This report was assembled with the assistance of Claude (Anthropic), which synthesized and reorganized the author's prior assignment deliverables into the Section 4.1 template; see the disclosure box on the title page. Per-assignment AI-use disclosures carried over from the source documents are consolidated below.

- **2.1 Designing a Data Product:** "Artificial intelligence tools were used to verify that the technical statements, parameter values, and requirement references made in this document are consistent with the background research on Europa ice-shell radar sounding, cryobot mission concepts, and planetary robotics requirements engineering. AI also generated fake data to demonstrate the data being recorded."
- **2.3 ConOps Sketch:** "AI was used to check grammar, cohesiveness and correct wording of the following."
- **3.4 Point-Mass Thermal Equilibrium:** "AI was used to fact-check the results."
- **3.5 Link Budget:** "I used AI to check the arithmetic behind each link budget term and to cross-check the assumed parameters against published Electra and DSN performance data. I picked the architecture and the parameters myself, and I set the margin the same way."
- **3.6 Mobility Solution:** "I used AI to check the arithmetic behind the traction, torque, and localization error budget numbers, and to cross-check the low-temperature ice friction values against published measurements. The mobility architecture is mine: the wheel and suspension configuration, and the call to run differential drive without steering actuators."

- **1.3 Deep Dive into a Robotic Mission:** "Claude Sonnet 4.6 (Anthropic) was used to compare this paper against the assignment rubric and provide a grade estimate."

3.2 Structures and Mechanisms Design (dedicated AI Use Disclosure): "The attached rover sketch was generated as an SVG graphic using Claude (model: Claude Fable 5), an AI assistant developed by Anthropic, accessed through the claude.ai interface. What the AI did: Produced the SVG drawing (linework, layout, labels, and leader lines) based on my direction. What the AI did not do: The design content — component selection, structural configuration, materials (Al 7075-T73 struts, Ti-6Al-4V vault), payload (REASON-heritage GPR), and requirements traceability — comes from my own prior coursework (requirements, trade study, and mass budget for this mission). The AI drew from those documents rather than originating the design. Process: The sketch went through several revision cycles that I directed, including corrections to the suspension architecture, GPR mounting orientation (nadir-pointing, belly-mounted), sensor mast structure, RTG mounting, and the landing strut separation mechanism. I reviewed the final output for accuracy against my mission design."